MECH 223 Solved Examples — Chapters 3 & 4

MECH 223-002 Engineering Statics, Fall 2026
Exam 2 review (Chapters 3 and 4)

Set A

How to use these. Try each example yourself after reading only *Given/Find* and the figure. Then compare with the Strategy and Solution. Sign convention: x right, y up, counterclockwise (CCW) moments positive. Numbers are new; problem ideas are noted where adapted.

1 Moment of a single force about a point

Level: easy. Chapter 3. Target confusion: a moment depends on ***which point*** *you take it about.*

1. Given / Find

A bracket is pinned to a wall at A. A cable pulls on the bracket at B, which is 0.400 m to the right of and 0.300 m above A. The cable force is $\mathbf{F} = (150\,\mathbf{i} - 200\,\mathbf{j})$ N.

Find: (a) the moment of $\mathbf{F}$ about A; (b) the moment of the same force about point C, which is 0.400 m to the right of A on the same horizontal line as A.

2. Figure

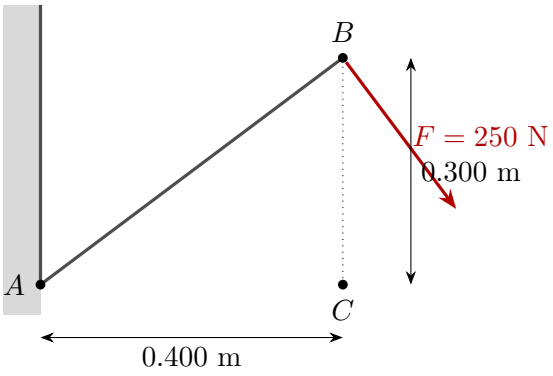


Figure 1: Bracket pinned to a wall at A. Point B is 0.400 m right and 0.300 m above A. A 250 N cable force acts at B, pointing right and down with components 150 N (right) and 200 N (down). Point C lies 0.400 m right of A on the line through A.

3. Free-body diagram

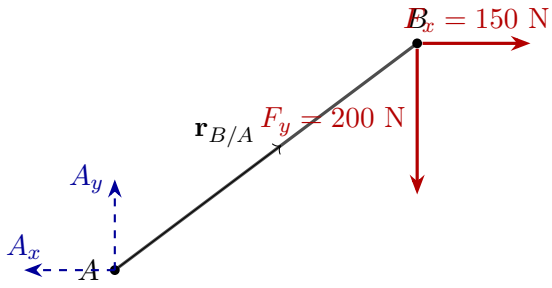


Figure 2: Free-body diagram of the bracket. Red solid arrows are the components of the cable force at B. Blue dashed arrows are the unknown pin reactions A_x and A_y at A. They pass through A, so they contribute zero moment about A. Position vector $\mathbf{r}_{B/A}$ runs from A to B.

4. Strategy

The moment of a force about a point is $\mathbf{M}_O = \mathbf{r} \times \mathbf{F}$, where $\mathbf{r}$ runs from the point to *any* point on the line of action of the force. In 2D this is $M = xF_y - yF_x$ with (x, y) the components of $\mathbf{r}$. We will compute it, then verify with $M = Fd$ using the perpendicular distance. Repeating with a second point shows the result changes with the point.

5. Solution

(a) About A. $\mathbf{r}_{B/A} = (0.400\,\mathbf{i} + 0.300\,\mathbf{j})$ m.

$$\begin{aligned} M_A &= xF_y - yF_x = (0.400)(-200) - (0.300)(150) \\ &= -80.0 - 45.0 = -125 \text{ N·m}. \end{aligned}$$

The negative sign means clockwise.

Check with $M = Fd$. The unit vector along $\mathbf{F}$ is $\mathbf{u} = (0.6, -0.8)$. The perpendicular distance from A to the line of action is $d = |\mathbf{r} \times \mathbf{u}| = |(0.400)(-0.8) - (0.300)(0.6)| = 0.500$ m. Then $|M_A| = Fd = (250)(0.500) = 125$ N·m. ✓

(b) About C. $\mathbf{r}_{B/C} = (0\,\mathbf{i} + 0.300\,\mathbf{j})$ m.

$$M_C = (0)(-200) - (0.300)(150) = -45.0 \text{ N·m}.$$

Here only F_x contributes, since F_y acts along the vertical line through C.

6. Answer

$\boxed{M_A = 125 \text{ N·m clockwise}}$ $\qquad$ $\boxed{M_C = 45.0 \text{ N·m clockwise}}$

7. Sanity check

The same force gives two different moments because the moment arm changed (0.500 m about A, 0.180 m about C, since $45.0/250 = 0.180$). Both are clockwise, which matches the picture: the force pushes B down and to the right, rotating the bracket clockwise about either point.

Inspired by the standard "moment of a force about a point" problems in Beer & Johnston, Ch. 3, with new numbers.

2 Couple versus moment of a force

Level: easy–medium. Chapter 3. Target confusion: the moment of a force depends on the point; the moment of a couple does not.

1. Given / Find

Two vertical forces act on a plate. $\mathbf{F}_1 = +120\ \text{N}\,\mathbf{j}$ acts at $D = (0.600, 0.200)$ m and $\mathbf{F}_2 = -120\ \text{N}\,\mathbf{j}$ acts at $E = (0.150, 0.350)$ m, with coordinates measured from the origin O. Let $G = (0.300, -0.100)$ m be another reference point.

Find: (a) the net force and net moment of the pair about O; (b) the net moment of the pair about G; (c) the moment of $\mathbf{F}_1$ *alone* about O and about G.

2. Figure

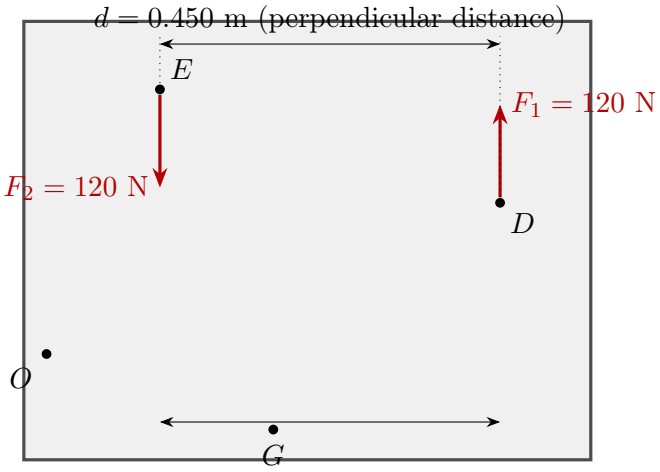


Figure 3: A plate with two parallel, equal, opposite vertical forces of 120 N each. F_1 acts upward at D and F_2 acts downward at E. The two lines of action are 0.450 m apart horizontally. O is the origin and G is a second reference point below the plate center.

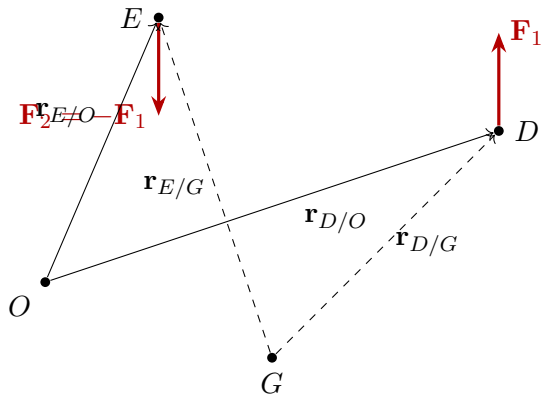


Figure 4: Free-body diagram of the plate showing only the two applied forces $\mathbf{F}_1$ and $\mathbf{F}_2 = -\mathbf{F}_1$ and the position vectors used for each reference point. Solid thin arrows run from O to D and to E. Dashed thin arrows run from G to D and to E.

3. Free-body diagram

4. Strategy

Compute $\sum \mathbf{M} = \sum \mathbf{r} \times \mathbf{F}$ about O and then about G, and compare. Because the forces are equal and opposite, the position vectors should combine into the single vector $\mathbf{r}_{D/E}$ and the reference point should drop out. Contrast this with one force alone, whose moment changes with the point.

5. Solution

(a) Net force: $\mathbf{F}_1 + \mathbf{F}_2 = \mathbf{0}$. About O, with $M = xF_y - yF_x$ and $F_x = 0$:

$$\begin{aligned} M_O &= x_D F_{1y} + x_E F_{2y} = (0.600)(120) + (0.150)(-120) \\ &= 72.0 - 18.0 = +54.0 \text{ N·m (CCW)}. \end{aligned}$$

(b) About G: $\mathbf{r}_{D/G} = (0.300,\, 0.300)$ m and $\mathbf{r}_{E/G} = (-0.150,\, 0.450)$ m.

$$M_G = (0.300)(120) + (-0.150)(-120) = 36.0 + 18.0 = +54.0 \text{ N·m}.$$

General reason. For any point P,

$$\mathbf{M}_P = \mathbf{r}_{D/P} \times \mathbf{F}_1 + \mathbf{r}_{E/P} \times (-\mathbf{F}_1) = (\mathbf{r}_{D/P} - \mathbf{r}_{E/P}) \times \mathbf{F}_1 = \mathbf{r}_{D/E} \times \mathbf{F}_1,$$

which does not contain P. Also $|M| = Fd = (120)(0.450) = 54.0$ N·m, with d the perpendicular distance between the two lines of action.

(c) Force $\mathbf{F}_1$ alone:

$$M_O^{(F_1)} = (0.600)(120) = +72.0 \text{ N·m}, \qquad M_G^{(F_1)} = (0.300)(120) = +36.0 \text{ N·m}.$$

	About O	About G
$\mathbf{F}_1$ alone	+72.0 N·m	+36.0 N·m
$\mathbf{F}_2$ alone	−18.0 N·m	+18.0 N·m
Pair (couple)	**+54.0** N·m	**+54.0** N·m

6. Answer

$$\boxed{\mathbf{F}_{\text{net}} = \mathbf{0}, \quad M_O = M_G = 54.0 \text{ N·m CCW}}$$

$\mathbf{F}_1$ alone: 72.0 N·m about O and 36.0 N·m about G (both CCW).

7. Sanity check

The couple moment equals force × perpendicular separation ($120 \times 0.450 = 54.0$ N·m) no matter where you compute it, so it is a free vector. A single force has no such property; its moment changed from 72.0 to 36.0 N·m when only the reference point moved.

Concept adapted from the couple discussion in Beer & Johnston, Ch. 3 ("couples"), with new numbers.

3 Equivalent force–couple system

Level: medium. Chapter 3. HW6-style equivalent system, new numbers.

1. Given / Find

A horizontal bar has A at its left end. Loads act on it as follows (positions measured from A):

- $\mathbf{F}_1 = (0, -300)$ N at (0.500, 0) m;
- $\mathbf{F}_2 = (160, -120)$ N at (1.20, 0.300) m (on a short bracket above the bar);
- $\mathbf{F}_3 = (0, +180)$ N at (2.00, 0) m;
- a pure couple $C = 90.0$ N·m, clockwise.

Find: (a) the equivalent force–couple system at A; (b) the equivalent system at B, the right end at (2.00, 0); (c) the point on the bar where the resultant force alone, with no couple, is equivalent to the original system.

2. Figure

3. Free-body diagram

4. Strategy

Two systems are equivalent if they have the same resultant force and the same total moment about one point (and hence about all points). So: sum the forces to get $\mathbf{R}$; sum the moments of every force *and* the couple about A to get M_A. Couples add directly to the moment sum without needing a position vector. To move the system to another point, use $\mathbf{M}_B = \mathbf{M}_A + \mathbf{r}_{A/B} \times \mathbf{R}$.

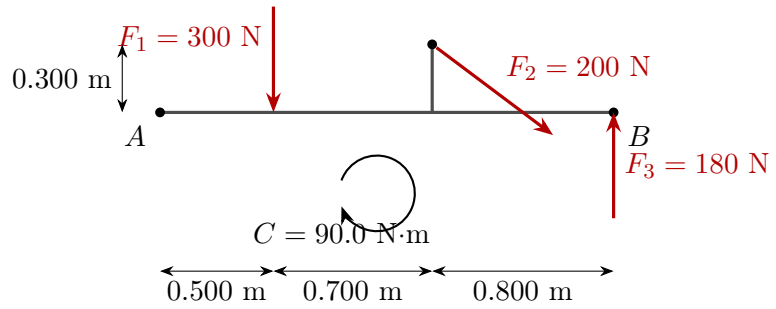


Figure 5: Horizontal bar from A (left) to B (right, 2.00 m from A). $F_1 = 300$ N acts downward 0.500 m from A. $F_2 = 200$ N (components 160 N right, 120 N down) acts at the top of a bracket 1.20 m from A and 0.300 m above the bar. $F_3 = 180$ N acts upward at B. A 90.0 N·m clockwise couple also acts on the bar. Horizontal dimensions are 0.500, 0.700 and 0.800 m in sequence.

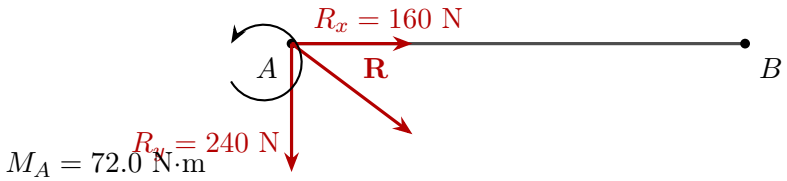


Figure 6: Equivalent system at A. The resultant force $\mathbf{R}$ (components $R_x = 160$ N right and $R_y = 240$ N down, magnitude 288 N) acts at A, together with a clockwise couple moment $M_A = 72.0$ N·m. This single force plus couple has the same total force and the same moment about every point as the four original loads.

5. Solution

Resultant force.

$$R_x = 0 + 160 + 0 = 160 \text{ N},$$
$$R_y = -300 - 120 + 180 = -240 \text{ N},$$
$$R = \sqrt{160^2 + 240^2} = 288 \text{ N}, \qquad \theta = \tan^{-1}\frac{240}{160} = 56.3^\circ \text{ below the } +x\text{-axis}.$$

Moment about A ($M = xF_y - yF_x$, CCW positive):

$$M_1 = (0.500)(-300) - 0 = -150 \text{ N·m},$$
$$M_2 = (1.20)(-120) - (0.300)(160) = -144 - 48.0 = -192 \text{ N·m},$$
$$M_3 = (2.00)(180) - 0 = +360 \text{ N·m},$$
$$M_C = -90.0 \text{ N·m} \quad \text{(couple, clockwise)}.$$

$$M_A = -150 - 192 + 360 - 90.0 = -72.0 \text{ N·m} \quad \text{(clockwise)}.$$

(b) Move to B. $\mathbf{r}_{A/B} = (-2.00,\, 0)$ m, so $\mathbf{r}_{A/B} \times \mathbf{R} = (-2.00)(-240) - 0 = +480$ N·m:

$$M_B = -72.0 + 480 = +408 \text{ N·m (CCW)}, \qquad \mathbf{R} \text{ unchanged}.$$

(c) Single force with no couple. Place $\mathbf{R}$ at $(x, 0)$ on the bar so its moment about A equals M_A:

$$x\,R_y = M_A \;\Rightarrow\; x(-240) = -72.0 \;\Rightarrow\; x = 0.300 \text{ m}.$$

6. Answer

$$\boxed{\text{At } A: \ \mathbf{R} = (160\,\mathbf{i} - 240\,\mathbf{j}) \text{ N } (288 \text{ N at } 56.3^\circ \text{ below } +x), \quad M_A = 72.0 \text{ N·m clockwise}}$$

$$\boxed{\text{At } B: \ \mathbf{R} \text{ same}, \ M_B = 408 \text{ N·m CCW}} \qquad \boxed{\text{Single resultant: } \mathbf{R} \text{ acting at } x = 0.300 \text{ m from } A}$$

7. Sanity check

Placing $\mathbf{R}$ alone at $x = 0.300$ m gives moment $0.300 \times (-240) = -72.0$ N·m about A, matching M_A, and moment $(0.300 - 2.00)(-240) = +408$ N·m about B, matching part (b). The two descriptions agree at both points, as equivalent systems must.

HW6-style equivalent system (replace several forces and a couple by a force at a chosen point plus a couple), with new geometry and numbers.

4 Simply supported beam: pin, roller, a load and a couple

Level: medium. Chapter 4. Target confusion: ***why*** *a pin gives two reaction components and a roller only one, and how the moment center removes unknowns.*

1. Given / Find

A horizontal beam AB is 4.00 m long. It is held by a **pin at** A and a **roller at** B that rests on a horizontal surface. A 500 N force with components 300 N to the right and 400 N downward acts on the beam 1.50 m from A. A 400 N·m counterclockwise couple is applied to the beam at 3.00 m from A. Treat the beam weight as negligible.

Find: the support reactions at A and B.

2. Figure

3. Free-body diagram

4. Strategy

Reactions come from what the support blocks.

- *Pin at A.* It stops the beam point from moving in *any* direction in the plane, so it can push or pull both horizontally and vertically: two unknowns, A_x and A_y.
- *Roller at B.* It stops motion only into or away from the surface it sits on, and lets the beam slide along it. The surface is horizontal, so the reaction is vertical only: one unknown, B_y. There is no horizontal reaction at B.

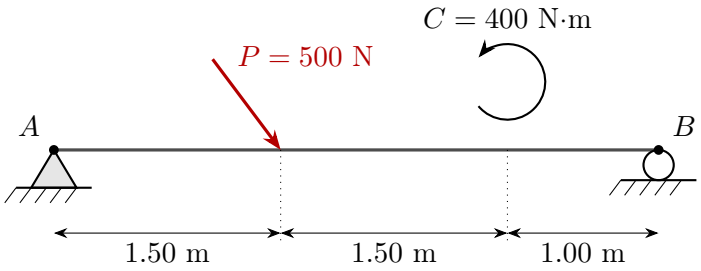


Figure 7: Beam AB, 4.00 m long, on a pin at A (triangle on the left) and a roller at B (circle on the right, free to roll horizontally). The 500 N force acts 1.50 m from A and points right and down, with components 300 N and 400 N. A 400 N·m counterclockwise couple acts on the beam 3.00 m from A. The horizontal dimensions are 1.50, 1.50 and 1.00 m in sequence.

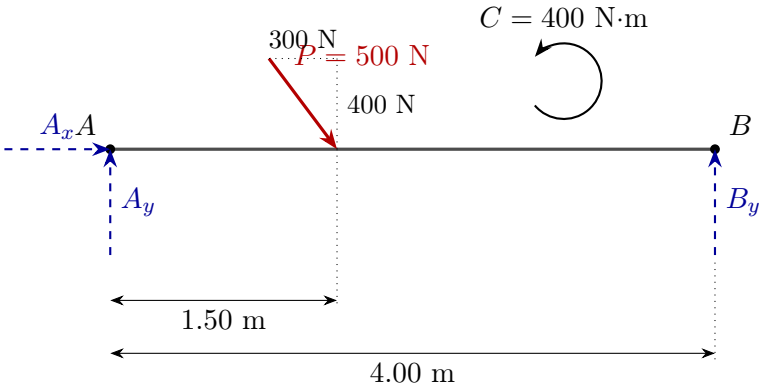


Figure 8: Free-body diagram of the beam alone. The supports are removed and replaced by their reactions (blue dashed arrows, drawn in their *assumed* positive directions): A_x to the right and A_y upward at the pin, and only a vertical B_y at the roller. The solution will show that A_x comes out negative, meaning it really points left. The applied 500 N force is shown with its components, and the couple is shown as a curved arrow.

That is three unknowns for three planar equations ($\sum F_x = 0$, $\sum F_y = 0$, $\sum M = 0$), so the beam is statically determinate.

Choose the moment center to kill unknowns. Taking moments about A removes A_x and A_y, since both pass through A. That leaves B_y as the only unknown in one equation. The 300 N component acts along the beam axis through A, so it has no moment about A. The couple enters the moment sum as its full value +400 N·m, wherever on the beam it is applied (a couple is a free vector, as in the earlier couple example). Then use the force equations for A_x and A_y, and check with a moment sum about B.

5. Solution

Moments about A (CCW positive; only the vertical 400 N component, B_y and the couple contribute):

$$\sum M_A = B_y(4.00) - (400)(1.50) + 400 = 0$$
$$4.00\,B_y = 600 - 400 = 200 \;\Rightarrow\; B_y = +50.0 \text{ N}.$$

Horizontal forces:

$$\sum F_x = A_x + 300 = 0 \;\Rightarrow\; A_x = -300 \text{ N}.$$

The negative sign means the pin pushes the beam to the *left*.

Vertical forces:

$$\sum F_y = A_y + B_y - 400 = 0 \;\Rightarrow\; A_y = 400 - 50.0 = +350 \text{ N}.$$

Pin reaction as one vector: $A = \sqrt{300^2 + 350^2} = 461$ N, directed $\tan^{-1}(350/300) = 49.4^\circ$ above the $-x$ direction (up and to the left).

Independent check, moments about B**.** Now B_y drops out. With r measured from B:

$$\sum M_B = -A_y(4.00) + (400)(2.50) + 400 = -1400 + 1000 + 400 = 0. \;\checkmark$$

6. Answer

$$\boxed{A_x = 300 \text{ N (to the left)}}, \qquad \boxed{A_y = 350 \text{ N (up)}}, \qquad \boxed{B_y = 50.0 \text{ N (up)}}$$

Equivalently the pin force is 461 N at 49.4° above the $-x$ axis. There is no horizontal force at the roller.

7. Sanity check

Net horizontal: the pin's 300 N to the left cancels the applied 300 N to the right. Net vertical: $350 + 50.0 = 400$ N up balances the 400 N down. Physically, the downward load alone would give a 600 N·m clockwise moment about A that B_y must resist, but the counterclockwise couple already supplies 400 N·m of it, which is why B_y is small. Because $B_y > 0$ the roller is being pushed on, as a roller requires. If a calculation had produced a negative B_y, the roller would have to *pull* on the beam, which it cannot do, and the beam would lift off.

Concept adapted from the standard simply supported beam problems in Beer & Johnston, Ch. 4, with new geometry, an added couple, and new numbers.

5 L-shaped bracket with a pin and a wall roller

Level: medium–hard. Chapter 4. Target confusion: the free-body-diagram workflow when the supports are of ***different types*** *and the roller is on a* ***vertical*** *surface.*

1. Given / Find

An L-shaped bracket is mounted against a vertical wall. Its vertical arm is 1.20 m tall and its horizontal arm sticks out 1.50 m from the wall. The bracket is held by a **pin at** A at the top of the vertical arm and a **roller at** C at the bottom corner, where the wheel rolls on the wall. A 500 N force acts at the end D of the horizontal arm, with components 300 N to the right (away from the wall) and 400 N downward. Neglect the bracket weight.

Find: all the support reactions at A and C.

2. Figure

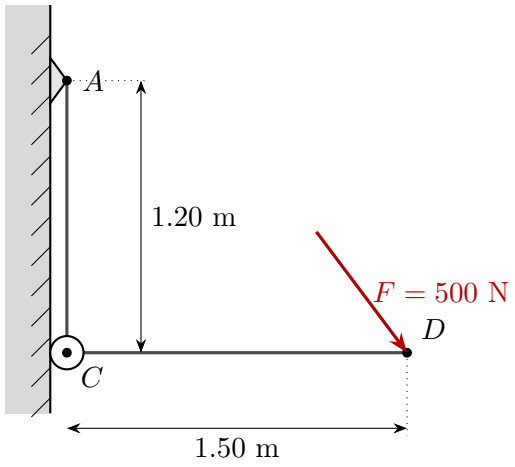


Figure 9: L-shaped bracket against a vertical wall (gray, left). The pin A joins the top of the vertical arm to the wall. The roller C is at the bottom corner, a wheel touching the wall. The vertical arm is 1.20 m long and the horizontal arm 1.50 m. A 500 N force acts at the free end D, pointing right and down with components 300 N (away from the wall) and 400 N (down).

3. Free-body diagram

4. Strategy

A reliable workflow for any support mix:

1. **Isolate** the bracket and sketch only the bracket.
2. **Replace each support by what it blocks.** Pin $\rightarrow$ two components (A_x, A_y). Roller on a vertical wall $\rightarrow$ one horizontal force (C_x), because the wheel is free to roll up and down the wall but cannot move into it or away from it.
3. **Draw the known load** with its components, and count: three unknowns (A_x, A_y, C_x), three equations.
4. **Moment center at the pin** A**.** Both A_x and A_y pass through A, so $\sum M_A = 0$ contains only C_x and the applied force.

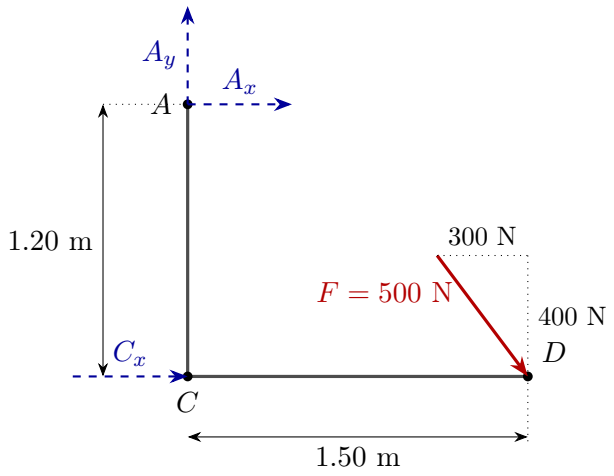


Figure 10: Free-body diagram of the bracket. The wall is removed. The pin at A becomes two components A_x (assumed to the right) and A_y (assumed up). The roller at C rolls on a *vertical* wall, so it can only push horizontally: a single C_x, assumed to the right (away from the wall, the only direction a roller can push). The applied force is shown with its components.

5. **Force sums** give A_x and A_y.
6. **Check** with a moment sum about a different point, here C.

Use $M = xF_y - yF_x$ with the position vector measured *from the moment center* to the point where the force acts. Place the origin at C: then $A = (0,\ 1.20)$ m, $C = (0,\ 0)$ and $D = (1.50,\ 0)$ m.

5. Solution

Moments about A. Here $\mathbf{r}_{C/A} = (0,\ -1.20)$ m and $\mathbf{r}_{D/A} = (1.50,\ -1.20)$ m.

$$M_{C_x} = xF_y - yF_x = (0)(0) - (-1.20)(C_x) = +1.20\,C_x,$$
$$M_F = (1.50)(-400) - (-1.20)(300) = -600 + 360 = -240 \text{ N·m}.$$

$$\sum M_A = 1.20\,C_x - 240 = 0 \;\Rightarrow\; C_x = +200 \text{ N}.$$

The positive sign agrees with the assumed direction: the wheel pushes away from the wall, as it must.

Horizontal and vertical forces:

$$\sum F_x = A_x + C_x + 300 = 0 \;\Rightarrow\; A_x = -200 - 300 = -500 \text{ N},$$
$$\sum F_y = A_y - 400 = 0 \;\Rightarrow\; A_y = +400 \text{ N}.$$

The pin pulls the bracket *toward* the wall with 500 N and holds it up with 400 N. As one vector, $A = \sqrt{500^2 + 400^2} = 640$ N, directed $\tan^{-1}(400/500) = 38.7°$ above the $-x$ direction.

Independent check, moments about C. With $\mathbf{r}_{A/C} = (0,\ 1.20)$ m and $\mathbf{r}_{D/C} = (1.50,\ 0)$ m:

$$\sum M_C = \underbrace{-(1.20)(A_x)}_{+600} + \underbrace{(0)(A_y)}_{0} + \underbrace{(1.50)(-400)}_{-600} = 0. \;\checkmark$$

6. Answer

$$\boxed{C_x = 200 \text{ N (to the right, away from the wall)}}$$

$$\boxed{A_x = 500 \text{ N (to the left)}}, \qquad \boxed{A_y = 400 \text{ N (up)}} \quad (A = 640 \text{ N at } 38.7^\circ \text{ above } -x).$$

7. Sanity check

The horizontal forces balance: $-500 + 200 + 300 = 0$. The 500 N force at D tries to rotate the bracket clockwise about the wall, pulling the top away from the wall and pushing the bottom in. The wall's reactions oppose this: the pin pulls the top in and the roller pushes the bottom out. Taken about A, the applied force has a net 240 N·m clockwise moment, and $C_x = 200$ N acting 1.20 m below A supplies the 240 N·m counterclockwise moment that balances it. Also, C_x had to be positive; a negative value would have meant the wheel was being pulled into the wall, which a roller cannot do.

Mixed-support bracket setup adapted from the common pin-plus-roller frame problems in Beer & Johnston, Ch. 4; geometry, loading and numbers are new.

6 Rigid beam on a pin and an inclined roller

Level: medium–hard. Chapter 4. Target confusion: the direction of a reaction is set by ***what the support blocks****, not by whatever direction looks convenient.*

1. Given / Find

A rigid horizontal beam AB is 3.00 m long. It is held by a **pin at** A and by a **frictionless roller at** B that rests on a smooth incline. The incline rises 3 units for every 4 horizontal units ($\theta = 36.9^\circ$ above the horizontal), going up toward the right. The beam carries a 800 N weight acting at 1.20 m from A (the load, with the beam's own weight included).

Find: the reaction at the roller B (magnitude and direction) and the pin reaction components at A.

2. Figure

3. Free-body diagram

4. Strategy

What each support blocks.

- *Pin at A:* blocks translation in both directions, so two components A_x, A_y.
- *Roller at B on a smooth incline:* blocks motion *into* the incline and nothing along it, because the wheel rolls freely along the surface and there is no friction. So the reaction is purely

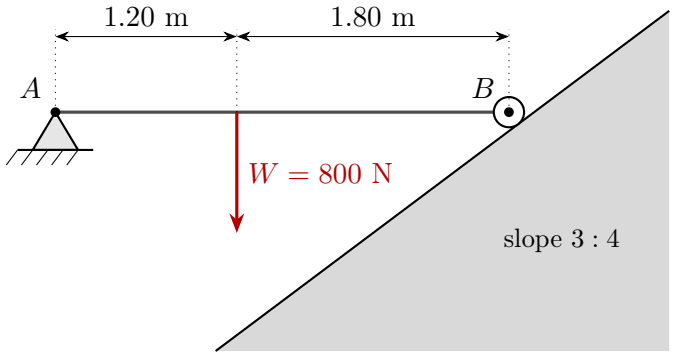


Figure 11: Horizontal beam AB, 3.00 m long, pinned at A (left) and resting on a roller at B (right). The roller sits on a smooth incline (gray) that rises 3 units vertically per 4 horizontally, so the surface makes an angle of 36.9° with the horizontal. An 800 N weight acts downward 1.20 m from A, and B is 1.80 m beyond it.

> **perpendicular to the incline**: one unknown magnitude N_B along a known direction. It is not vertical, and it has a horizontal component.

The incline direction is $(4, 3)/5 = (0.8,\ 0.6)$. A perpendicular unit vector is $(-3, 4)/5 = (-0.6,\ 0.8)$, chosen to point away from the solid incline (a roller can push, not pull). Thus

$$\mathbf{N}_B = N_B\,(-0.6\,\mathbf{i} + 0.8\,\mathbf{j}).$$

The unknowns are A_x, A_y, N_B: three unknowns, three equations. Moments about A remove A_x and A_y and leave only N_B, so solve for N_B first, then use the force sums.

5. Solution

Moments about A. B is at $\mathbf{r}_{B/A} = (3.00,\ 0)$ m. The force N_B has components $(-0.6N_B,\ 0.8N_B)$. Its horizontal component acts along the beam line and has no moment; only the vertical component does. The weight acts at $x = 1.20$ m.

$$\sum M_A = (3.00)(0.8\,N_B) - (1.20)(800) = 0$$
$$2.40\,N_B = 960 \;\Rightarrow\; N_B = 400 \text{ N}.$$

So the components at B are $N_{Bx} = -0.6(400) = -240$ N and $N_{By} = 0.8(400) = +320$ N.

Force sums:

$$\sum F_x = A_x + N_{Bx} = A_x - 240 = 0 \;\Rightarrow\; A_x = +240 \text{ N},$$
$$\sum F_y = A_y + N_{By} - 800 = A_y + 320 - 800 = 0 \;\Rightarrow\; A_y = +480 \text{ N}.$$

The pin force is $A = \sqrt{240^2 + 480^2} = 537$ N at $\tan^{-1}(480/240) = 63.4°$ above the $+x$ axis.

Independent check, moments about B. Now A_x and N_B drop out (their lines pass through B or along the beam):

$$\sum M_B = -(3.00)(A_y) + (1.80)(800) = -1440 + 1440 = 0. \;\checkmark$$

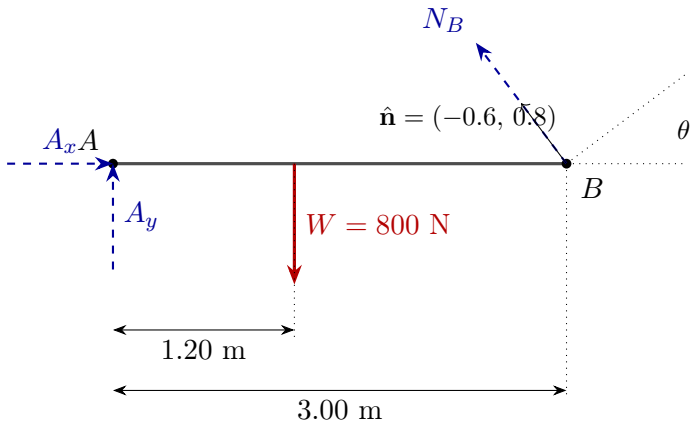


Figure 12: Free-body diagram of the beam. At the pin: assumed components A_x (right) and A_y (up). At the roller: a *single* force N_B perpendicular to the incline, pointing away from it (up and to the left). Its direction is the unit normal $\hat{\mathbf{n}} = (-0.6,\ 0.8)$, which is perpendicular to the incline direction $(0.8,\ 0.6)$. The angle $\theta = 36.9°$ between the incline and the horizontal is shown with the dotted lines.

What if the roller is drawn vertical? If we pretended B sat on a flat floor, we would take B_y vertical: $\sum M_A$ gives $3.00B_y = 960$, so $B_y = 320$ N and $\sum F_x$ gives $A_x = 0$. Both are wrong. The incline pushes the beam toward the *left* with 240 N, and the pin must push back to the right with 240 N.

6. Answer

$\boxed{N_B = 400 \text{ N, perpendicular to the incline, up and to the left}}$ $(N_{Bx} = -240 \text{ N},\ N_{By} = +320 \text{ N})$

$\boxed{A_x = 240 \text{ N (to the right)}}$, $\boxed{A_y = 480 \text{ N (up)}}$ $(A = 537 \text{ N at } 63.4° \text{ above } +x)$.

The normal direction makes an angle of 36.9° with the vertical, the same as the incline angle with the horizontal.

7. Sanity check

Vertical: $480 + 320 = 800$ N, matching the weight. Horizontal: the incline's 240 N push to the left is cancelled by the pin's 240 N to the right. Reasonableness: the weight is 40% of the way along the beam, so a straight-up support at B would carry about 40% of 800, i.e. 320 N, and the vertical component of N_B is exactly that. The extra horizontal part arises purely because the roller's reaction is tilted by the incline. As required, $N_B > 0$ (pushing away from the incline surface).

Idea of a roller on an inclined surface from standard rigid-body support problems in Beer & Johnston, Ch. 4 ("reaction normal to the surface"); the geometry and numbers are new.

7 Boom with a pulley, a rope and a support cable

Level: exam-level. Chapter 4. Target confusion: a rope over an ideal pulley pulls with the ***same tension on both sides****, and when the pulley is bolted to the body,* ***both rope segments*** *act on that body.*

1. Given / Find

A horizontal boom AB is 4.00 m long and is pinned to a wall at A. Its own weight is negligible. A frictionless pulley of negligible radius is mounted at the free end B. A rope is tied to the wall at point E, which is 3.00 m below A. The rope runs up to the pulley at B, passes over it, and carries a hanging weight $W = 450$ N on its other end, which hangs at rest. A separate support cable runs from point D on the boom, 3.00 m from A, to a wall point C that is 4.00 m above A.

Find: (a) the tension T in the support cable DC; (b) the pin reaction at A (components, magnitude and direction).

2. Figure

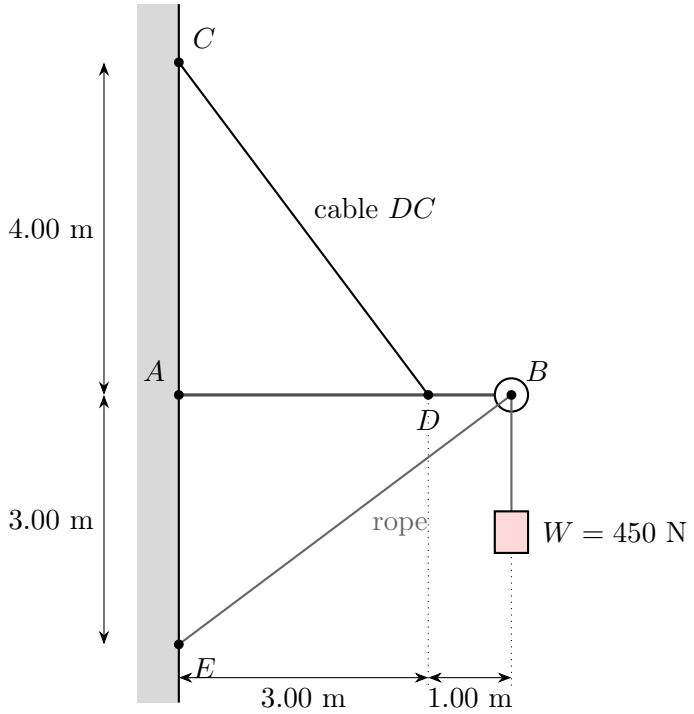


Figure 13: A horizontal boom AB, 4.00 m long, pinned to a wall at A (left, gray). A frictionless pulley is mounted at the free end B. A rope is tied to the wall at E, 3.00 m below A, runs up to the pulley, passes over it, and hangs straight down carrying a 450 N weight. A support cable joins D on the boom (3.00 m from A) to C on the wall, 4.00 m above A.

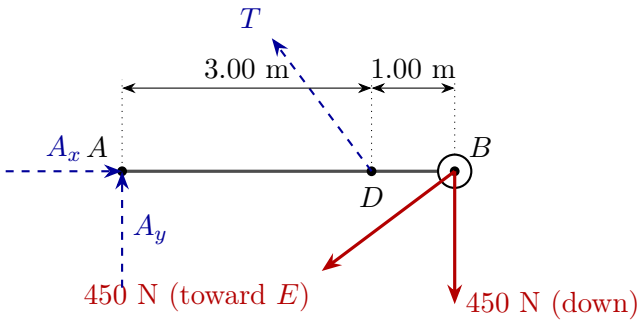


Figure 14: Free-body diagram of the boom together with its pulley. The cable at D gives one force T along the line DC (up and toward the wall). Because the pulley is part of the body, *both* rope segments touching it are drawn: one pulls straight down with 450 N (the hanging side) and the other pulls toward the anchor E (down and toward the wall) with the same 450 N. Blue dashed arrows are the unknowns T, A_x and A_y.

3. Free-body diagram

4. Strategy

Ideal pulley. The hanging weight is at rest, so the rope tension on that side equals $W = 450$ N. A frictionless pulley changes the rope's direction but not its tension, so the segment running to E also carries 450 N. This is a *known* force, not an unknown.

Choice of body. Take the boom and pulley as one rigid body. The rope touches the pulley on two sides, so two rope forces act at B. Each pulls *along the rope, away from the body*. The support cable pulls along DC. The pin gives A_x and A_y. The unknowns are T, A_x, A_y: three unknowns, three equations.

Directions. From $D(3, 0)$ to $C(0, 4)$ the vector is $(-3, 4)$, length 5, so the cable direction is $(-0.6,\ 0.8)$. From $B(4, 0)$ to $E(0, -3)$ the vector is $(-4, -3)$, length 5, so the rope direction is $(-0.8,\ -0.6)$.

Order. Moments about A eliminate A_x and A_y and give T at once. Then the force sums give A_x and A_y.

5. Solution

Forces at the pulley. Hanging segment: $(0,\ -450)$ N. Segment toward E: $450\,(-0.8,\ -0.6) = (-360,\ -270)$ N. Both act at B.

Moments about A (use $M = xF_y - yF_x$; every force acts on the line $y = 0$, so only vertical components matter):

$$\sum M_A = (3.00)(0.8\,T) + (4.00)(-270) + (4.00)(-450) = 0$$
$$2.40\,T = 1080 + 1800 = 2880 \;\Rightarrow\; T = 1200 \text{ N}.$$

Force sums. The cable force is $T(-0.6,\ 0.8) = (-720,\ +960)$ N.

$$\sum F_x = A_x - 720 - 360 = 0 \ \Rightarrow\ A_x = +1080 \text{ N},$$
$$\sum F_y = A_y + 960 - 270 - 450 = 0 \ \Rightarrow\ A_y = -240 \text{ N}.$$

The negative sign means A_y acts *downward*, opposite to the arrow assumed on the FBD. Then

$$A = \sqrt{1080^2 + 240^2} = 1106 \text{ N}, \qquad \tan^{-1}\left(\frac{240}{1080}\right) = 12.5^\circ \text{ below the } +x \text{ axis}.$$

Independent check, moments about B. Now A_x and the three forces at B drop out, leaving A_y at lever arm 4.00 m (to the left) and T at 1.00 m (to the left):

$$\sum M_B = (-4.00)(A_y) + (-1.00)(960) = (-4.00)(-240) - 960 = 960 - 960 = 0. \ \checkmark$$

Common mistake: leaving out the rope to E. If only the weight is drawn at B, the moment equation becomes $2.40\,T = 1800$, giving $T = 750$ N, which is 450 N too low. The rope runs over the pulley and *also* pulls on the body, so both segments belong on the FBD.

6. Answer

$$\boxed{T = 1200 \text{ N}}$$

$\boxed{A_x = 1080 \text{ N (to the right)}}$, $\boxed{A_y = 240 \text{ N (downward)}}$ $(A = 1106 \text{ N at } 12.5^\circ \text{ below } +x)$.

7. Sanity check

Vertical balance: $-240 + 960 - 270 - 450 = 0$. Horizontal balance: $1080 - 720 - 360 = 0$. The moment check about B used different unknowns and also closed. Reasonableness: the downward pull at B totals $270+450 = 720$ N at a 4.00 m arm, while the cable acts at only 3.00 m, so its vertical component must be $720 \times 4/3 = 960$ N; dividing by the direction factor 0.8 gives $T = 1200$ N, as found. A_y points down because the cable's upward pull (960 N) exceeds the 720 N of downward rope pull, so the pin must pull the boom down by the difference, 240 N.

The idea of a rope over a pulley mounted on a pinned boom, with an ideal-pulley tension rule, is standard in the rigid-body equilibrium chapters of Beer & Johnston and Hibbeler; the geometry and numbers are new.

8 Two-force and three-force members

Level: medium. Chapter 4. Target confusion: a member loaded at ***exactly two points*** *carries a force along the line joining them, and a member loaded at* ***exactly three points*** *has force lines that meet at one point.*

1. Given / Find

Part (a), two-force member. A horizontal beam AD is pinned to a wall at A. A straight link BC is pinned to the beam at B and pinned to the wall at C. The link is weightless and carries no load between its pins. The beam is 5.00 m long. Point B is 3.00 m from A along the beam, and C is 4.00 m directly below A. A load $P = 600$ N acts downward at D. **Find:** the force in link BC (say whether it is in tension or compression) and the pin reaction at A.

Part (b), three-force member. A uniform bar AB is 4.00 m long and weighs 600 N. It is pinned to a wall at A and held horizontal by a cable attached at B and tied to the wall at point C, which is 3.00 m above A. **Find:** the point where the three force lines of action meet, the direction of the pin force at A, and the magnitudes of the cable tension and the pin force.

2. Figure

Part (a).

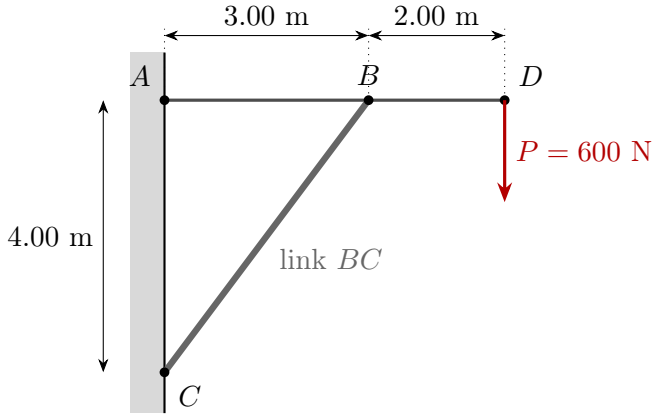


Figure 15: Part (a). Beam AD (5.00 m) is pinned to the wall at A (left, gray). Link BC is pinned to the beam at B, 3.00 m from A, and to the wall at C, 4.00 m below A, so the link rises 3 to the right for every 4 up. A 600 N load pulls straight down at D, 2.00 m beyond B.

Part (b).

3. Free-body diagram

Part (a). The beam is drawn on the left and the link on the right. The force on the link at each end is the equal-and-opposite partner of the force on whatever it is pinned to.

Part (b). The dotted lines extend the lines of action of the weight and the cable; they cross at point P. The pin reaction must pass through that same point.

4. Strategy

Two-force member. If a rigid body is in equilibrium with forces at only two points, the two forces must be equal in magnitude, opposite in direction and collinear. The only line that joins the

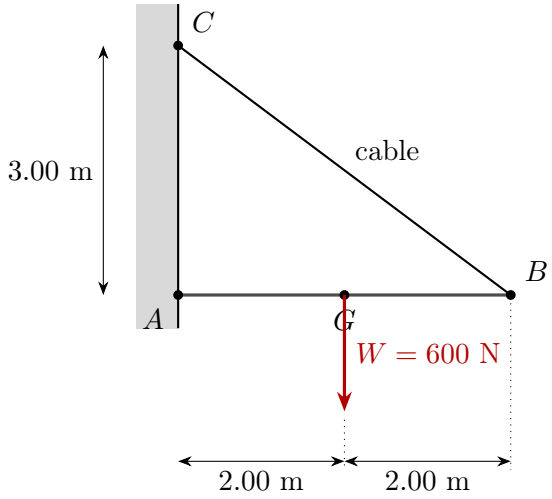


Figure 16: Part (b). A uniform bar AB, 4.00 m long, is pinned to the wall at A and held horizontal by a cable from its end B to the wall at C, 3.00 m above A. The 600 N weight acts at the midpoint G, 2.00 m from each end.

two points is the line through them. So a straight, weightless link that is pinned at both ends and carries no load in between has a force along its own axis, with magnitude F at both ends. That removes the four pin-component unknowns at B and C and replaces them with one number.

Three-force member. Suppose a body is in equilibrium under three forces whose lines of action are not all parallel. Take the point where two of the lines intersect. The moment of those two forces about that point is zero, so the moment of the third must also be zero. A nonzero force has zero moment about a point only if its line of action passes through that point. Hence the three lines are *concurrent*. (If two of the forces are parallel, the third must be parallel to them too, since the three forces sum to zero.) Concurrency gives the *direction* of an unknown force for free.

Plan. In (a), use the link direction $(3, 4)/5 = (0.6, 0.8)$ to write the link force on the beam with a single unknown, then take moments about A. In (b), find the point P, take the pin force along AP, and close the force triangle.

5. Solution

Part (a). The force the link exerts on the beam at B lies along BC. Assume tension, so the link pulls the beam toward C, direction $(-0.6, -0.8)$: components $(-0.6F, -0.8F)$.

Moments about A (only the vertical components have lever arms along the beam):

$$\sum M_A = (3.00)(-0.8F) + (5.00)(-600) = 0$$
$$-2.40F = 3000 \;\Rightarrow\; F = -1250 \text{ N}.$$

The negative sign means the link is in **compression**: it pushes the beam at B with 1250 N in the direction (0.6, 0.8), that is, (+750, +1000) N.

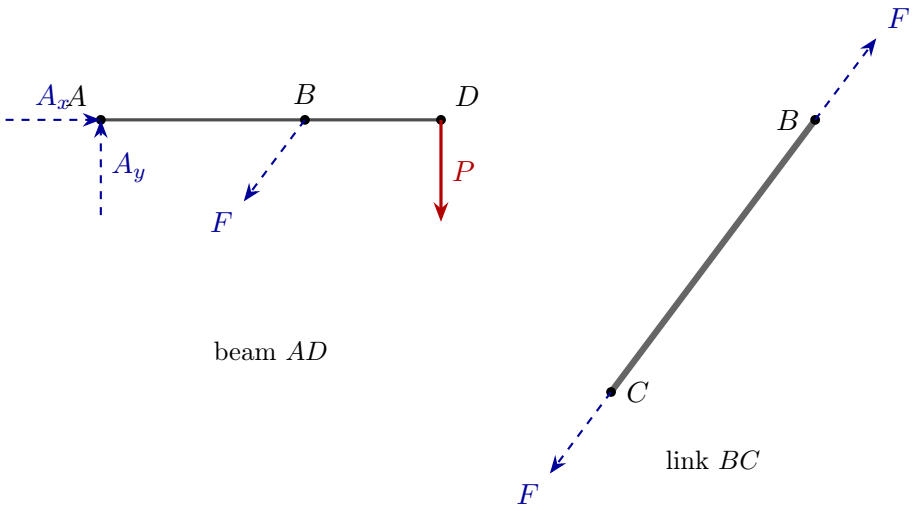


Figure 17: Part (a), free-body diagrams. Left: the beam AD with the 600 N load P at D, the assumed pin components A_x and A_y at A, and one force F at B along the link direction, drawn as if the link were in tension (pulling the beam down and toward the wall). Right: the link BC with only two forces on it, both of magnitude F, equal and opposite and along the line BC, drawn pulling outward at each end.

Force sums on the beam:

$$\sum F_x = A_x + 750 = 0 \;\Rightarrow\; A_x = -750 \text{ N},$$
$$\sum F_y = A_y + 1000 - 600 = 0 \;\Rightarrow\; A_y = -400 \text{ N}.$$

The pin force is $A = \sqrt{750^2 + 400^2} = 850$ N, pointing left and down at $\tan^{-1}(400/750) = 28.1°$ below the $-x$ axis.

Check, moments about B: $(-3.00)(A_y) + (2.00)(-600) = (-3.00)(-400) - 1200 = 0$. ✓

Without the two-force idea, the link would need its own FBD with pin components at both B and C: six unknowns and six equations. With it, the beam alone has three unknowns and three equations.

Part (b). The weight acts at the midpoint, so its line is the vertical $x = 2.00$ m. The cable line passes through $B(4, 0)$ and $C(0, 3)$. At $x = 2.00$ m it is at height $y = \frac{3}{4}(4 - 2) = 1.50$ m. So the lines meet at $P = (2.00,\ 1.50)$ m.

The line of the pin force passes through $A(0, 0)$ and P, so its direction is $(2.00, 1.50)/2.50 = (0.8, 0.6)$, which is 36.9° above the $+x$ axis. The force must point up and to the right (toward P) to balance the weight, so $\mathbf{R}_A = R\,(0.8, 0.6)$ with $R > 0$. The cable force on the bar is $\mathbf{T} = T(-0.8, 0.6)$ and the weight is $(0, -600)$ N.

$$\sum F_x = 0.8R - 0.8T = 0 \;\Rightarrow\; R = T,$$
$$\sum F_y = 0.6R + 0.6T - 600 = 0 \;\Rightarrow\; 1.2\,T = 600 \;\Rightarrow\; T = R = 500 \text{ N}.$$

The components are $R_x = 400$ N (right) and $R_y = 300$ N (up).

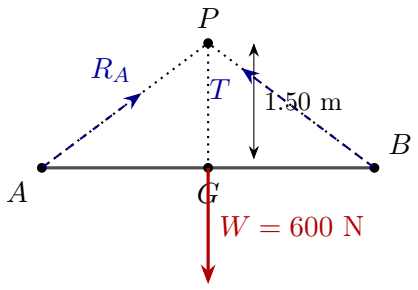


Figure 18: Part (b), free-body diagram of the bar. The three forces are the weight W (vertical, through G), the cable tension T at B (along BC, up and toward the wall) and the pin reaction R_A at A (direction unknown at first). Dotted lines extend the lines of action of W and T; they cross at P, which lies 2.00 m from the wall and 1.50 m above the bar. The line of R_A must also pass through P.

Independent check, moments about A: $(4.00)(0.6\times 500)+(2.00)(-600)=1200-1200=0$. ✓ The pin force passes through A and contributes nothing.

Why concurrency helped. A pin normally has two unknown components, so the bar has three unknowns. Concurrency fixes the pin force's direction, leaving two unknown magnitudes, and the two force-sum equations settle them without a moment equation.

6. Answer

Part (a).

$\boxed{F_{BC} = 1250 \text{ N (compression)}}$, $\boxed{A = 850 \text{ N}}$ ($A_x = 750$ N left, $A_y = 400$ N down; 28.1° below the $-x$ axis).

Part (b).

$\boxed{T = 500 \text{ N}}$, $\boxed{R_A = 500 \text{ N at } 36.9^\circ \text{ above the } +x \text{ axis}}$ ($R_x = 400$ N right, $R_y = 300$ N up),

with the three lines of action meeting at $P = (2.00, 1.50)$ m.

7. Sanity check

(a) Vertical balance on the beam: $-400+1000-600=0$. Horizontal: $-750+750=0$. Notice also that the beam in (a) is itself a three-force member (pin, link, load). The link force line through B in the direction (0.6, 0.8) meets the load line $x = 5.00$ m at $y = \frac{4}{3}(2.00) = 2.67$ m, and the line from A to (5.00, 2.67) has slope $0.533 = 400/750$, the same as the pin force found above. The concurrency test agrees with the algebra. Reasonableness: the load is 5.00 m out and the link's vertical push acts at only 3.00 m, so the link force's vertical part must exceed 600 N, and 1000 N does.

(b) Vertical: $300 + 300 = 600$ N matches the weight. Horizontal: 400 right at A and 400 left from the cable. Reasonableness: the cable and pin force make equal angles with the vertical, which is why $T = R$, and each carries half the weight in the vertical direction. Both have positive values, consistent with the assumed senses.

The two-force and three-force member tests are standard material in Beer & Johnston and Hibbeler, Ch. 4; the frame, bar, geometry and numbers here are new.

9 Bent bar with a pin, a link, a couple and two forces

*Level: exam-level. Chapter 4. Target confusion: with **several loads** on one body, the FBD must show* every *one of them, a couple adds its moment **with no lever arm**, and a pinned two-force link counts as **one** unknown, not two.*

1. Given / Find

A bent bar ABC has a horizontal leg AB of length 4.00 m and a vertical leg BC of length 3.00 m hanging down from B. Its weight is negligible. The bar is pinned to a wall at A. A straight, weightless link CE is pinned to the bar at C and pinned to the same wall at E, which is 6.00 m below A. Three loads act on the bar: a downward force of 800 N at the midpoint G of AB, a horizontal force of 500 N pointing right at C, and a couple of magnitude 1300 N·m acting counterclockwise on the bar.

Find: (a) the force in link CE, and whether it is in tension or compression; (b) the pin reaction at A (components, magnitude and direction).

2. Figure

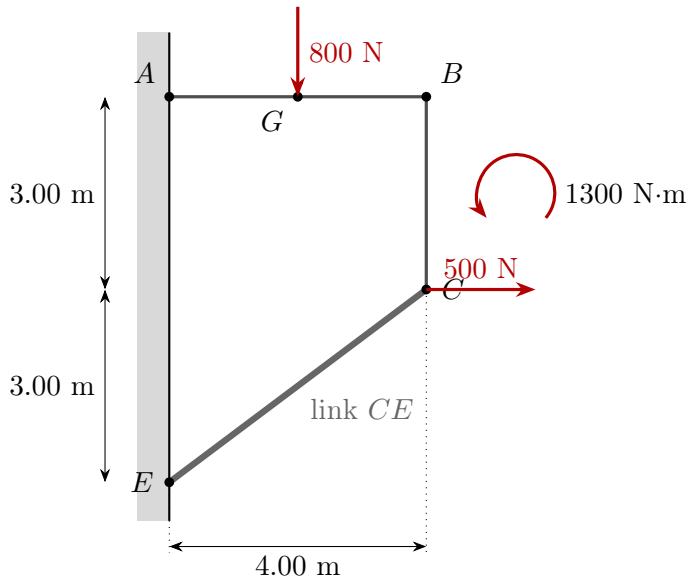


Figure 19: Bent bar ABC pinned to the wall at A (left, gray). Leg AB is 4.00 m long and leg BC drops 3.00 m. Link CE joins C to the wall pin at E, 6.00 m below A. Loads: 800 N down at the midpoint G of AB; 500 N to the right at C; a 1300 N·m couple, shown as a counterclockwise arc, acting on the bar.

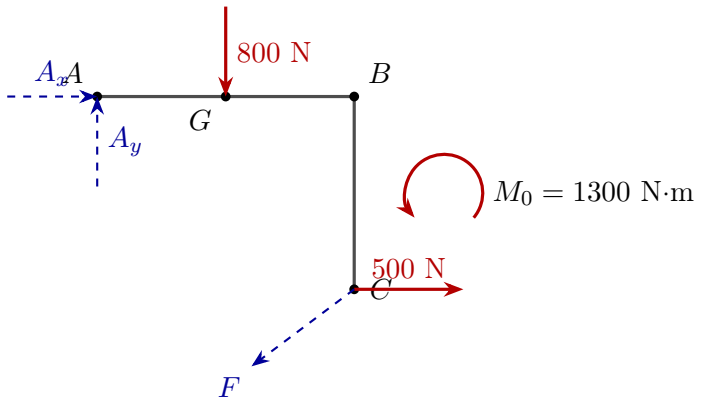


Figure 20: Free-body diagram of the bar alone. Red solid arrows are the known loads, including the couple M_0 (a free moment, not tied to any point). Blue dashed arrows are the unknowns: the pin components A_x, A_y, and the link force F at C, drawn as if the link were in tension (pulling the bar down and toward the wall). The link force lies along CE, which falls 3 for every 4 it runs to the left.

3. Free-body diagram

4. Strategy

Link. Link CE is weightless, loaded only at its two pins, so it is a two-force member. Its force on the bar acts along CE. From $C(4, -3)$ to $E(0, -6)$ the vector is $(-4, -3)$, length 5, so the unit vector is $(-0.8,\ -0.6)$. The link adds *one* unknown, F.

Body and unknowns. Take the bent bar alone. The unknowns are A_x, A_y and F: three unknowns, and the planar problem has three equations ($\sum F_x$, $\sum F_y$, $\sum M$).

Moment center. Take moments about A. Both pin components pass through A and drop out, so one equation gives F. Then the two force sums give A_x and A_y.

Couple. A couple contributes the same moment about every point, equal to its own magnitude, with sign from its rotation sense. It appears in $\sum M$ but *never* in $\sum F_x$ or $\sum F_y$.

Moment of each force about A ($M = xF_y - yF_x$, with (x, y) measured from A, CCW positive): the link force at $C(4, -3)$ has components $(-0.8F,\ -0.6F)$; the 800 N load acts at $(2, 0)$; the 500 N force acts at $(4, -3)$.

5. Solution

Moments about A:

$$\sum M_A = \underbrace{(4)(-0.6F) - (-3)(-0.8F)}_{\text{link force}} + \underbrace{(2)(-800)}_{\text{800 N load}} + \underbrace{-(-3)(500)}_{\text{500 N force}} + \underbrace{1300}_{\text{couple}} = 0$$

$$= -4.80F - 1600 + 1500 + 1300 = 0$$

$$4.80F = 1200 \ \Rightarrow \ F = 250 \text{ N}.$$

F is positive, so the link is in **tension**, as assumed. Its force on the bar is $250(-0.8,\ -0.6) = (-200,\ -150)$ N.

Force sums:

$$\sum F_x = A_x - 200 + 500 = 0 \;\Rightarrow\; A_x = -300 \text{ N},$$
$$\sum F_y = A_y - 150 - 800 = 0 \;\Rightarrow\; A_y = +950 \text{ N}.$$

The negative A_x means the pin pushes the bar to the *left*. Then

$$A = \sqrt{300^2 + 950^2} = 996 \text{ N}, \qquad \tan^{-1}\left(\frac{950}{300}\right) = 72.5^\circ \text{ above the } -x \text{ axis}.$$

Independent check, moments about C. The link force drops out. Measured from C, point A is at $(-4, +3)$ and the 800 N load acts at $(-2, +3)$. The 500 N force acts at C itself:

$$\sum M_C = \underbrace{(-4)(950) - (3)(-300)}_{\text{pin at } A} + \underbrace{(-2)(-800)}_{800 \text{ N}} + 0_{(500 \text{ N at } C)} + 1300$$
$$= -3800 + 900 + 1600 + 1300 = 0. \;\checkmark$$

Common mistakes. (i) Multiplying the couple by a distance. It is already a moment of 1300 N·m. (ii) Drawing the link with pin components at both C and E: that adds three unneeded unknowns and an unsolvable system. (iii) Dropping the 500 N force from $\sum M_A$. Its line of action is 3.00 m *below* A, and a rightward push there turns the bar counterclockwise, contributing +1500 N·m.

6. Answer

$$\boxed{F_{CE} = 250 \text{ N (tension)}}$$

$\boxed{A_x = 300 \text{ N (to the left)}}$, $\boxed{A_y = 950 \text{ N (upward)}}$ ($A = 996$ N at 72.5° above the $-x$ axis).

7. Sanity check

Horizontal balance: $-300 - 200 + 500 = 0$. Vertical balance: $950 - 150 - 800 = 0$. The moment check about C used a different set of unknowns and also closed. Reasonableness: the link pulls the bar *down and toward the wall*, so A_y must exceed the 800 N load, and 950 N does, by exactly the 150 N vertical part of the link force. The pin pushes left because the 500 N push to the right is larger than the 200 N pull toward the wall from the link. Finally, about A the counterclockwise moments are the couple and the 500 N force ($1300 + 1500 = 2800$ N·m) and the clockwise ones are the 800 N load and the link ($1600 + 1200 = 2800$ N·m), so they balance.

A bent bar held by a pin and a two-force link, with a couple and several loads, is a standard rigid-body equilibrium arrangement in Beer & Johnston and Hibbeler, Ch. 4; the geometry, loads and numbers here are new.

10 3D equilibrium: a plate on a ball-and-socket and three cables

*Level: exam-level. Chapter 4. Target confusion: in 3D there are **six** equations and a cable force must be written as a **magnitude times a unit vector**; a moment about a well-chosen **axis** removes several unknowns at once.*

1. Given / Find

A thin, rigid, rectangular plate $ABCD$ lies horizontally. With origin at A and axes x (along AB), y (along AD) and z (vertically up), the corners are $A(0,0,0)$, $B(4,0,0)$, $C(4,3,0)$ and $D(0,3,0)$, in meters. The weight of the plate is negligible. The plate is held by a ball-and-socket joint at A and by three cables:

- cable BG is vertical, from B up to $G(4,0,3)$;
- cable DE is vertical, from D up to $E(0,3,3)$;
- cable CE runs from C up to the same anchor $E(0,3,3)$.

A force $\mathbf{P} = (600\,\mathbf{i} - 900\,\mathbf{k})$ N acts on the plate at the point $(3,2,0)$ m.

Find: the three cable tensions T_B, T_D, T_C and the reaction at A in vector form.

2. Figure

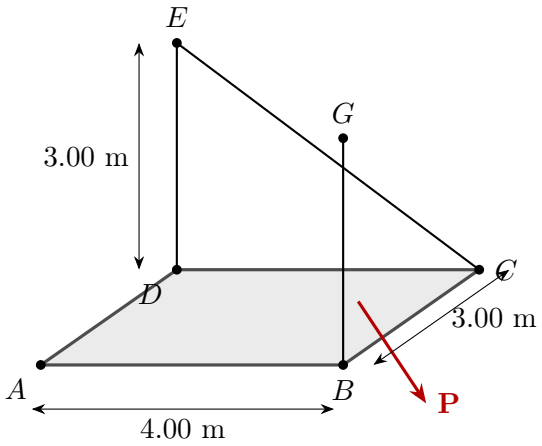


Figure 21: Oblique view. The plate $ABCD$ is horizontal; x runs to the right, y runs away from the viewer (drawn up and to the right), and z is up, a right-handed set. The ball-and-socket is at A. Cables BG and DE are vertical (3.00 m long). Cable CE goes from C to the same anchor E as cable DE: 4.00 m toward the $-x$ direction and 3.00 m up. The force $\mathbf{P}$ acts at $(3,2,0)$ m with components 600 N in $+x$ and 900 N in $-z$.

3. Free-body diagram

4. Strategy

Unknowns and equations. A ball-and-socket gives three force components and each cable gives one tension, so there are six unknowns. In 3D, $\sum \mathbf{F} = \mathbf{0}$ and $\sum \mathbf{M} = \mathbf{0}$ give six scalar equations. Six equations, six unknowns.

Cable direction. Write each cable force as (tension)$\times$(unit vector from the plate toward the anchor):

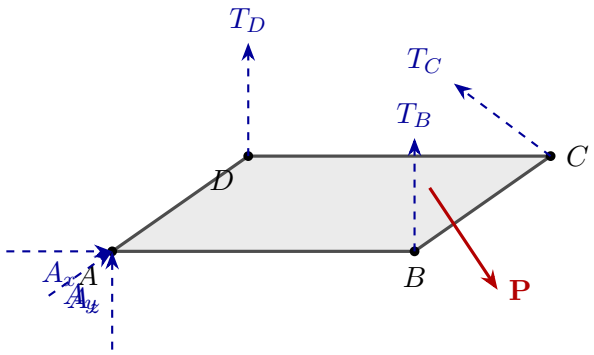


Figure 22: Free-body diagram of the plate. Red: the known force $\mathbf{P}$. Blue dashed: the six unknowns. At A the ball-and-socket supplies three components A_x, A_y, A_z (drawn in their positive senses). Each cable can only pull, so T_B, T_D, T_C are drawn toward their anchors: straight up at B and D, and up and toward the $-x$ direction at C along CE.

- BG and DE: unit vector $\mathbf{k}$;
- CE: from $C(4,3,0)$ to $E(0,3,3)$ the vector is $(-4,0,3)$, length 5, so the unit vector is $(-0.8,\ 0,\ 0.6)$.

Moment center and axes. Take moments about A with $\mathbf{M}_A = \sum \mathbf{r}\times\mathbf{F} = \mathbf{0}$. The ball-and-socket force passes through A, so none of A_x, A_y, A_z appear. Each component of this vector equation is a moment about an axis through A. Choosing the axis well removes more unknowns:

- the z-axis through A: a vertical force has no moment about a vertical axis, so the two vertical cables drop out and only T_C and $\mathbf{P}$ remain;
- the x-axis is the line AB. It passes through *two* unknown forces, the ball-and-socket at A and the cable at B, so the moments of A_x, A_y, A_z and T_B about it are all zero, leaving only T_D and T_C.

The force sums then give the ball-and-socket reaction.

5. Solution

Position vectors from A (m): $\mathbf{r}_B = 4\mathbf{i}$, $\mathbf{r}_C = 4\mathbf{i}+3\mathbf{j}$, $\mathbf{r}_D = 3\mathbf{j}$, $\mathbf{r}_P = 3\mathbf{i}+2\mathbf{j}$.

Forces on the plate (N): $\mathbf{T}_B = T_B\mathbf{k}$, $\mathbf{T}_D = T_D\mathbf{k}$, $\mathbf{T}_C = T_C(-0.8\,\mathbf{i}+0.6\,\mathbf{k})$, $\mathbf{P} = 600\,\mathbf{i}-900\,\mathbf{k}$, and $\mathbf{A} = A_x\mathbf{i}+A_y\mathbf{j}+A_z\mathbf{k}$.

Moment of each force about A (using $\mathbf{i}\times\mathbf{j}=\mathbf{k}$, $\mathbf{j}\times\mathbf{k}=\mathbf{i}$, $\mathbf{k}\times\mathbf{i}=\mathbf{j}$, so $\mathbf{i}\times\mathbf{k}=-\mathbf{j}$):

$$
\begin{aligned}
\mathbf{r}_B\times\mathbf{T}_B &= (4\mathbf{i})\times(T_B\mathbf{k}) = -4T_B\,\mathbf{j},\\
\mathbf{r}_D\times\mathbf{T}_D &= (3\mathbf{j})\times(T_D\mathbf{k}) = 3T_D\,\mathbf{i},\\
\mathbf{r}_C\times\mathbf{T}_C &= (4\mathbf{i}+3\mathbf{j})\times T_C(-0.8\,\mathbf{i}+0.6\,\mathbf{k}) = T_C\,(1.8\,\mathbf{i}-2.4\,\mathbf{j}+2.4\,\mathbf{k}),\\
\mathbf{r}_P\times\mathbf{P} &= (3\mathbf{i}+2\mathbf{j})\times(600\,\mathbf{i}-900\,\mathbf{k}) = -1800\,\mathbf{i}+2700\,\mathbf{j}-1200\,\mathbf{k}.
\end{aligned}
$$

Adding them and setting each component of $\sum \mathbf{M}_A$ to zero:

$$\begin{aligned}
\sum M_z: &\quad 2.4\,T_C - 1200 = 0 && \Rightarrow\ T_C = 500 \text{ N},\\
\sum M_x: &\quad 3\,T_D + 1.8\,T_C - 1800 = 0 && \Rightarrow\ 3T_D = 1800 - 900 \ \Rightarrow\ T_D = 300 \text{ N},\\
\sum M_y: &\quad -4\,T_B - 2.4\,T_C + 2700 = 0 && \Rightarrow\ 4T_B = 2700 - 1200 \ \Rightarrow\ T_B = 375 \text{ N}.
\end{aligned}$$

The $\sum M_x$ line above is the moment about the axis AB through two unknowns (A and B). Notice that A_x, A_y, A_z and T_B never appear in it.

Force sums. With $\mathbf{T}_C = 500(-0.8,\ 0,\ 0.6) = (-400,\ 0,\ 300)$ N:

$$\begin{aligned}
&\sum F_x = A_x - 400 + 600 = 0 \ \Rightarrow\ A_x = -200 \text{ N},\\
&\sum F_y = A_y = 0 \ \Rightarrow\ A_y = 0,\\
&\sum F_z = A_z + 375 + 300 + 300 - 900 = 0 \ \Rightarrow\ A_z = -75 \text{ N}.
\end{aligned}$$

So $\mathbf{A} = (-200\,\mathbf{i} - 75\,\mathbf{k})$ N, and $|\mathbf{A}| = \sqrt{200^2 + 75^2} = 214$ N.

Independent check, moment about the line BC (the axis through B parallel to y, which contains the points of application of T_B and T_C, so both drop out). Measured from B, A is at $(-4, 0, 0)$, D at $(-4, 3, 0)$ and the load point at $(-1, 2, 0)$. The y-components are

$$\begin{aligned}
\sum M_{BC} &= \underbrace{4A_z}_{\text{ball at } A} + \underbrace{4T_D}_{\text{cable } DE} + \underbrace{(-900)}_{\mathbf{P}}\\
&= 4(-75) + 4(300) - 900 = -300 + 1200 - 900 = 0. \ \checkmark
\end{aligned}$$

This equation used the reaction and a tension found from different equations, so it tests the whole solution.

Common mistakes. (i) Treating cable CE as vertical. It has a -0.8 horizontal part, and that part alone supplies the moment about z that fixes T_C. (ii) A sign slip in $\mathbf{i} \times \mathbf{k} = -\mathbf{j}$. (iii) Using 2D thinking: with only $\sum F_x$, $\sum F_y$ and one moment, the problem cannot be solved.

6. Answer

$$\boxed{T_B = 375 \text{ N}}, \qquad \boxed{T_D = 300 \text{ N}}, \qquad \boxed{T_C = 500 \text{ N}}$$

$$\boxed{\mathbf{A} = (-200\,\mathbf{i} + 0\,\mathbf{j} - 75\,\mathbf{k}) \text{ N}} \quad (|\mathbf{A}| = 214 \text{ N}:\ 200 \text{ N toward } -x,\ 75 \text{ N downward}).$$

7. Sanity check

All three tensions are positive, consistent with cables that can only pull. Vertical balance: the cables supply $375 + 300 + 0.6(500) = 975$ N upward against the 900 N downward part of $\mathbf{P}$, so the ball-and-socket must supply 75 N *downward*, matching $A_z = -75$ N. Horizontal balance: $\mathbf{P}$ pushes 600 N toward $+x$, cable CE pulls 400 N toward $-x$, and the socket supplies the remaining 200 N toward $-x$. No force has a y-component, so $A_y = 0$ is expected. The moment check about the line BC closed with a different combination of unknowns.

A horizontal plate held by a ball-and-socket and cables is a standard 3D rigid-body equilibrium arrangement in Beer & Johnston and Hibbeler, Ch. 4; the layout, load and numbers here are new.